

Affichage-Ar-HCN-polonais

June 19, 2026

1 Introduction

Dans ce Notebook, je travaille **uniquement** sur les données ab-initio de **Toczyłowski**.

J'ai seulement récupéré les paramètres obtenus avec les "meilleurs" fit des méthodes `curve_fit` et `least_squares`, qui, vous le verrez sont quasi identiques en terme de résultats.

J'ai ajouté les paramètres donnés par Toczyłowski dans sa thèse à titre de comparaison.

1.1 Tableau des Erreurs pas méthode :

Méthode	R ²	RMSE (mEh)	MAE (mEh)
Toczyłowski	0.9973571231491378	0.14053712510676145	0.07967955352389043
<code>curve_fit</code> + 'lm'	0.9999998091291987	0.00119432452372047	0.00087006388462275
<code>curve_fit</code> + 'lm' + sigma	0.9999910167165791	0.00819351310342679	0.00276581346507194
<code>curve_fit</code> + 'trf'	0.999997616897787	0.00133451646592886	0.00094146809737027
<code>curve_fit</code> + 'trf' + sigma	0.9999247424840818	0.02371524093658603	0.00688134668613111
<code>least_squares</code>	0.999997618394195	0.00133409741213665	0.00094175867130043
<code>least_squares</code> + 'jac'	0.999998091292266	0.00119432443657110	0.00087005663911204
<code>least_squares</code> + 'jac' + 'soft'	0.999998091292273	0.00119432443466397	0.00087005338244569
<code>least_squares</code> + 'jac' + 'huber'	0.999998091292276	0.00119432443330507	0.00087005649593775

Méthode	R ²	RMSE (cm ⁻¹)	MAE (cm ⁻¹)
Toczyłowski	0.9973571231491378	30.844322096260684	17.487605474609086
<code>curve_fit</code> + 'lm'	0.9999998091291987	0.2621241445163892	0.1909574902355770
<code>curve_fit</code> + 'lm' + sigma	0.9999910167165791	1.7982675553449088	0.6070264446832262
<code>curve_fit</code> + 'trf'	0.999997616897787	0.2928928987596549	0.2066283907162378
<code>curve_fit</code> + 'trf' + sigma	0.9999247424840818	5.204892897347624	1.5102786155597213
<code>least_squares</code>	0.999997618394195	0.2928009419314839	0.2066921462514732
<code>least_squares</code> + 'jac'	0.999998091292266	0.2621241253934157	0.1909559001545393
<code>least_squares</code> + 'jac' + 'soft'	0.999998091292273	0.2621241249748908	0.1909551855687354
<code>least_squares</code> + 'jac' + 'huber'	0.999998091292276	0.2621241246766935	0.1909558687387600

Les méthodes en **gras** sont celles pour lesquelles j'ai sélectionné les paramètres pour ce Notebook

Rappel :

- R2 : sans unité, indique sur la tendance globale
- RMSE : Racine de l'erreur quadratique moyenne (ici en cm^{-1}), mise en valeur des erreurs importantes
- MAE : Erreur absolue moyenne (ici en cm^{-1}), mise en valeur de l'erreur moyenne

2 Paramètres

2.1 Toczyłowski

- b1 = 1.48547
- b2 = 0.05926
- b3 = -0.07533
- b4 = 0.12892
- b5 = 0.04978
- b6 = -0.00004

2.1.1 c

- c1 = -134348.77
- c2 = -281277.47
- c3 = -57041.638
- c4 = -116961.41

2.1.2 d

- d1 = 14.23209
- d2 = 0.99248
- d3 = 0.88287
- d4 = 0.70551
- d5 = 0.38213
- d6 = 0.06818

2.1.3 g

- g11 = 0.06053, g12 = -0.12937, g13 = 0.07849, g14 = -0.05992, g15 = 0.04541, g16 = 0.03519
 - g21 = 0.00640, g22 = 0.04016, g23 = -0.02325, g24 = 0.02468, g25 = -0.02338, g26 = -0.01536
 - g31 = -0.00277, g32 = -0.00416, g33 = 0.00221, g34 = -0.00239, g35 = 0.00392, g36 = 0.00207
 - g41 = 0.00013, g42 = 0.00015, g43 = -0.00006, g44 = 0.00003, g45 = -0.00021, g46 = -0.00009
-

2.2 least_squares

2.2.1 b

- b1 = 1.483407536976
- b2 = 0.112015542111
- b3 = -0.032843612191

- $b4 = 0.094365726134$
- $b5 = 0.033820767387$
- $b6 = -0.004629355901$
- $c1 = -135783.5865435$
- $c2 = -325692.4344777$
- $c3 = -57995.31109569$
- $c4 = -125784.8469770$

2.2.2 d

- $d1 = 14.37087301922$
- $d2 = 1.210291512784$
- $d3 = 1.127352935061$
- $d4 = 0.466341968270$
- $d5 = 0.263961941831$
- $d6 = 0.043692710808$

2.2.3 g

- $g11 = 0.044277336958$, $g12 = -0.150928579110$, $g13 = 0.029017675752$, $g14 = -0.029401214750$, $g15 = 0.013423710578$, $g16 = 0.009201449094$
 - $g21 = 0.008911721077$, $g22 = 0.050508135234$, $g23 = -0.008855531861$, $g24 = 0.013531186127$, $g25 = -0.008965289979$, $g26 = -0.005064185661$
 - $g31 = -0.002917801252$, $g32 = -0.005428112556$, $g33 = 0.001146212702$, $g34 = -0.001178096509$, $g35 = 0.001834942387$, $g36 = 0.000752982787$
 - $g41 = 0.000137627196$, $g42 = 0.000192853374$, $g43 = -0.000053626009$, $g44 = -0.000008219267$, $g45 = -0.000113691774$, $g46 = -0.000031076263$
-

2.3 curve_fit (lm)

2.3.1 b

- $b1 = 1.483341392272$
- $b2 = 0.112108256596$
- $b3 = -0.032852332034$
- $b4 = 0.094353514677$
- $b5 = 0.033841179578$
- $b6 = -0.004631208182$

2.3.2 c

- $c1 = -135782.7803631$
- $c2 = -325700.0072445$
- $c3 = -57995.99546213$
- $c4 = -125793.2982190$

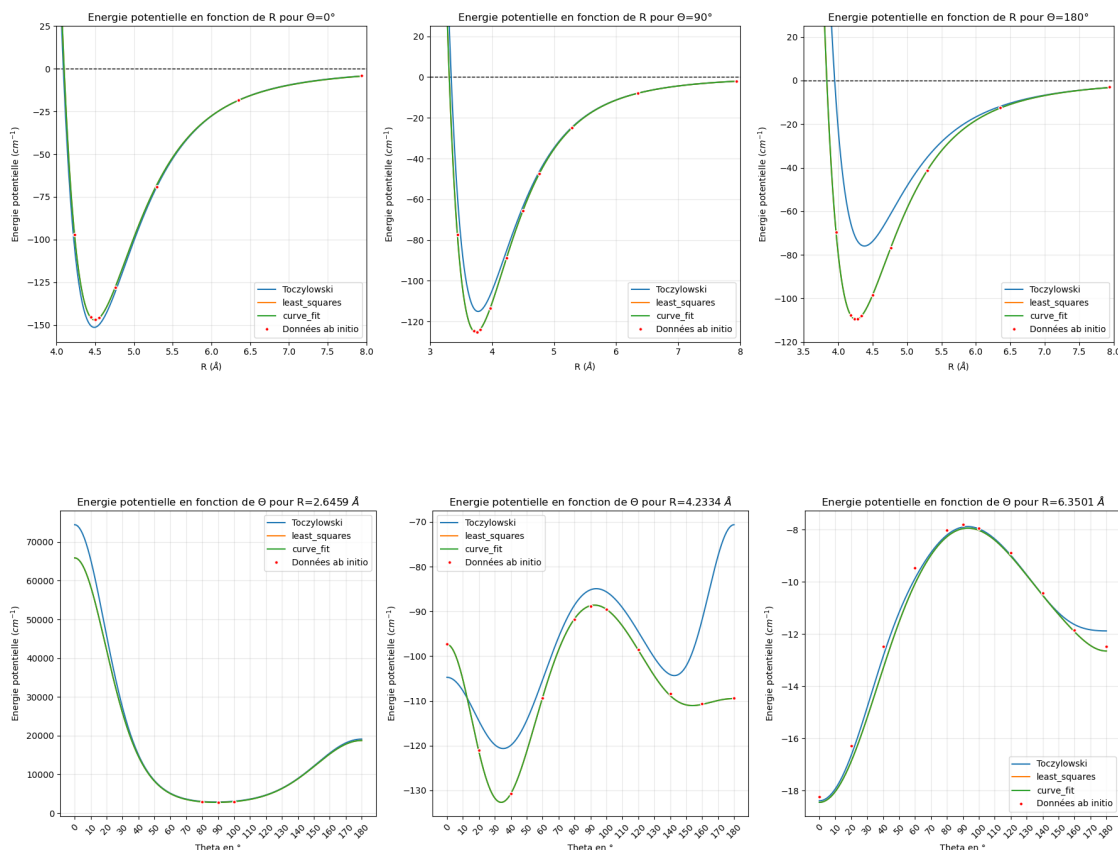
2.3.3 d

- $d1 = 14.35461101684$
- $d2 = 1.211144747196$
- $d3 = 1.127140847163$
- $d4 = 0.466324375937$
- $d5 = 0.264060522926$
- $d6 = 0.0436934169244$

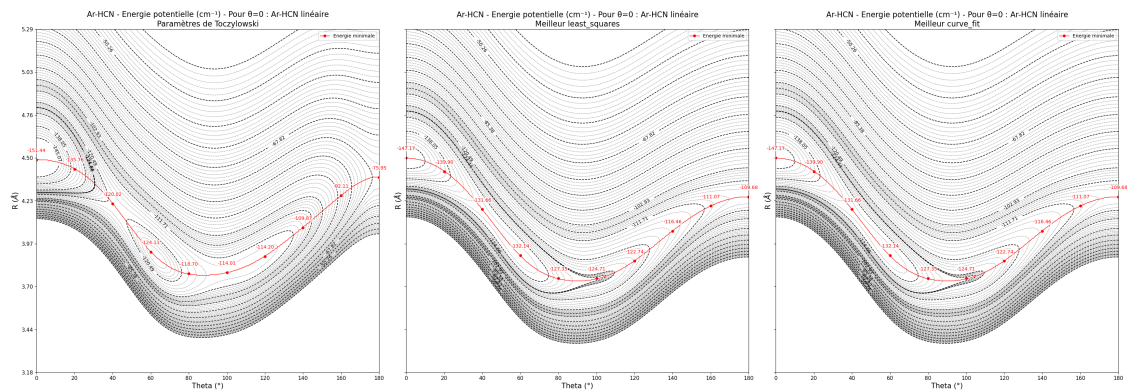
2.3.4 g

- $g11 = 0.045056225523$, $g12 = -0.153463196586$, $g13 = 0.029572096885$, $g14 = -0.029891893776$, $g15 = 0.013621363325$, $g16 = 0.0093643479828$
- $g21 = 0.009030278981$, $g22 = 0.051350634602$, $g23 = -0.009027915297$, $g24 = 0.013755264144$, $g25 = -0.009102520151$, $g26 = -0.0051522967751$
- $g31 = -0.002962013419$, $g32 = -0.005518046318$, $g33 = 0.001168114049$, $g34 = -0.001198159181$, $g35 = 0.001863583771$, $g36 = 0.0007660945489$
- $g41 = 0.000139756080$, $g42 = 0.000196022159$, $g43 = -0.000054625833$, $g44 = -0.000008278760$, $g45 = -0.000115484641$, $g46 = -0.0000316266424$

3 Courbe à valeurs fixées

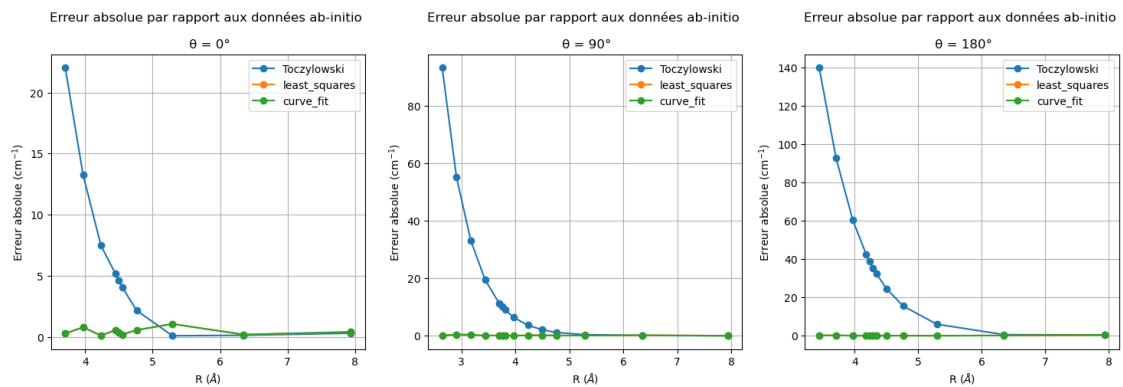
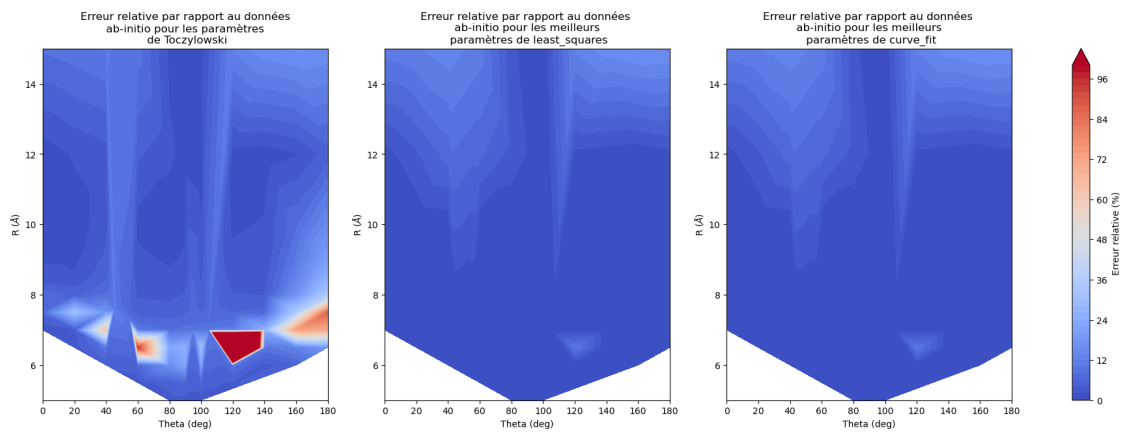


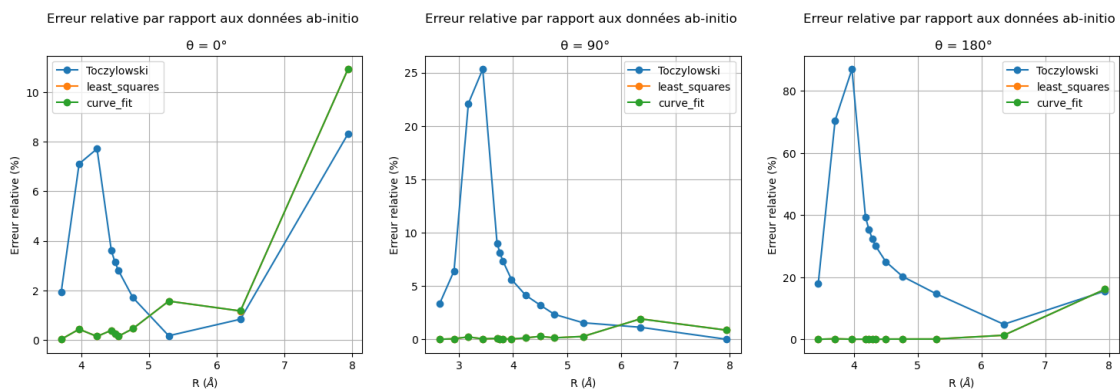
4 Contour plot



4.1 Erreur relative

[123]: `<matplotlib.colorbar.Colorbar at 0x7f295b33cd70>`





4.2 Minimum Energy Path

